skarm: A scikit-learn-Compatible Estimator for Association-Rule Mining with Auditable Pruning

Dudigam Ramya, Gontla Venkata Anvitha, Abhishek Kumar, Kamma Giridhar, and Paladagula Venkata Rajeev

Abstract—Frequent-itemset mining and association-rule mining are standard tools for market-basket analysis, recommendation, and categorical pattern discovery, yet current scikit-learn offers no native estimator for them: the feature request (issue #32712) was closed as "not planned" on scope grounds, with maintainers directing the work to a separate, scikit-learn-compatible library. The closest existing library, mlxtend, provides the mining functions but not an estimator API, and it provides no Eclat. We present skarm, a Python library that packages three classic miners — Apriori, FP-Growth, and Eclat — behind a single scikit-learn-style estimator with deterministic rule ordering, explicit candidate-count telemetry, and a bounded-generation mode that reports truncation instead of silently returning partial results. The evaluation protocol compares all three engines plus an mlxtend baseline at matched support thresholds on 25 datasets, with a 30-trial protocol on an 8-dataset subset and exact agreement checks on every cell. Across 25 datasets (1650 trial records on one Windows PC), all 157 compared engine pairs agree exactly on frequent itemsets; at each dataset's primary threshold, apriori fastest on none; fpgrowth fastest on 5 of 25; eclat fastest on 20 of 25, and mlxtend could not run 6 cells for stated reasons (3 dense-cap skips, 3 empty-input errors).

Index Terms—association rule mining, frequent itemset mining, Apriori, FP-Growth, Eclat, scikit-learn, auditable pruning

I. Introduction

A. Problem

Association-rule mining finds patterns of the form $X \Rightarrow Y$ over transactions, where X and Y are disjoint itemsets. The task is unsupervised: unlike classification, there is no target variable, and the output is a set of patterns whose size can explode combinatorially as the support threshold falls. Any library that offers this task must therefore answer three questions honestly: which patterns were found, whether the output is complete, and what it cost. Most teaching implementations answer none of them.

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B. The scikit-learn scope story, stated honestly

In November 2025 a community member asked scikit-learn for a native Apriori estimator (issue #32712 [7]), proposing a new sklearn.association module with fit() for frequent itemsets and rule generation with support, confidence, lift, and leverage. The maintainers closed the request on 16 November 2025 as "not planned", ruling that pattern mining sits outside the scope of machine learning as defined in scikit-learn, and suggested the work belonged in scikit-mine or scikit-learn-contrib. An earlier discussion (#2662, 2013–2014 [6]) had reached the same conclusion. skarm therefore does not claim to implement a requested scikit-learn feature, and it does not claim sklearn endorsement. It follows the maintainers' advice: a separate library that is compatible with scikit-learn's API conventions (estimator with fit, fitted attributes with a trailing underscore, parameter inspection), so it can sit beside scikit-learn pipelines rather than inside them.

C. Capstone objectives

This work is the KL University capstone project KLCAP-2026-00340. Its objectives, as registered in the portal specification, are: (O1) build an Apriori baseline; (O2) add FP-Growth and Eclat behind one estimator; (O3) make pruning auditable through candidate counts and a bounded mode; (O4) verify correctness against independent implementations; and (O5) evaluate on a broad, documented dataset collection. The portal contract additionally requires a held-out partition reserved before tuning (acceptance criterion AC-3), negative tests NT-1 to NT-6, and KPI evidence including a 30-trial statistical protocol (KPI-4) and an objective analytical oracle scoring rubric (KPI-2); Section V maps each requirement to its evidence.

D. Contributions

- One estimator, three engines: Apriori, FP-Growth, and Eclat with identical inputs, outputs, and rule metrics, so comparisons are between algorithms, not between APIs.
- Auditability by default: deterministic rule ordering, peak candidate counts, a degraded-mode flag with truncation counts, and recall against the exact result whenever bounded mode is used. Bounded output is never presented as exact.

- A documented 25-dataset evaluation collection with measured statistics and provenance for every file (Section IV), and a benchmark protocol (Section V) whose thresholds, trial counts, timeout rule, and correctness gate are fixed before the results are read.

II. Related work

A. Classic algorithms

Apriori [1] mines level by level, generating candidates from the previous level's frequent itemsets and pruning by anti-monotonicity (any superset of an infrequent itemset is discarded without counting). FP-Growth [2], [3] compresses the database into an FP-tree in two scans and mines it recursively without candidate generation. Eclat [4] uses a vertical layout (tidsets per item, support by intersection, equivalence classes traversed depth-first; diffsets are the dense-data refinement storing differences instead of full tidsets). All exact miners are exponential in the worst case; cost lands differently: Apriori in candidates plus one scan per level, FP-Growth in tree memory, Eclat in tidset memory. Secondary sources disagree on which engine wins on dense versus sparse data, so this paper asserts no universal winner and reports its own matched-threshold measurements instead.

B. Libraries and baselines

mlxtend [9] is the honest baseline. Its frequent_patterns module provides Apriori, FP-Growth, FP-Max, and H-Mine plus association_rules (support, confidence, lift, leverage, conviction, Zhang's metric), but as functions over a one-hot DataFrame, not estimators: no fit/get_params/clone, no Pipeline placement, no Eclat. Its documentation's toy timing (microseconds on a five-transaction example) is not a benchmark and is never quoted here as a speedup claim. scikit-mine [8] is the ecosystem home the sklearn maintainers pointed to; it offers SLIM, LCM, and LCMMax (closed, maximal, and MDL-compressed patterns) plus periodic and emerging-pattern miners, but not plain Apriori, FP-Growth, and Eclat with rule metrics, and its master branch showed no commits after May 2023 at the time of writing (to be re-checked at submission). SPMF [10] supplies the FIMI dataset collection and reference miners (including LCM-family algorithms); modern miners such as FIN, PrePost+, and LCMFreq [12] are stronger recent baselines and are acknowledged as future work in Section VII rather than silently omitted. A 2022 challenges survey [11] frames why pattern mining remains an open systems problem (scalability, interestingness, dynamic data) rather than a solved 1990s topic.

C. What skarm adds

skarm invents no mining algorithm. Its contribution is the package around the three classics: one estimator API, deterministic outputs, candidate telemetry, a bounded mode that reports its own truncation, and metric definitions with stated edge-case conventions (Section III). That package is what the sklearn issue asked for in API terms, and what mlxtend does not provide in estimator terms.

III. Methods

A. Definitions and rule metrics

A transaction database D contains N transactions over an item universe I. For an itemset X, $\mathrm{supp}(X)$ is the fraction of transactions containing X. A rule $X \Rightarrow Y$ (with $X \cap Y = \emptyset$) is evaluated at a minimum support threshold and a minimum confidence threshold. Table I defines the metrics skarm reports; the edge cases are part of the definition, not footnotes, because they change rule counts.

B. The three engines

Apriori generates candidates level by level and counts each level with a database scan, pruning by the anti-monotone property. FP-Growth builds an FP-tree ordered by descending item frequency, then mines conditional trees recursively; it performs no candidate generation but its memory is the tree plus conditional trees. The header table keeps a tail pointer per item, so node-link appends are $O(1)$ rather than walking each chain (added with the final two datasets; it changes no results, as Table V shows). Eclat intersects tidsets depth-first within equivalence classes, a decomposition originating in [5] and developed in [4]; its memory is the tidsets (or diffsets on dense data). All three engines consume the same pre-encoded sparse matrix and return the same fitted attributes, so a difference in output is a bug, not an API difference — which is exactly what the correctness gate in Section V checks.

C. Estimator API

AssociationRulesEstimator exposes min_support, metric (default confidence), min_threshold (default 0.25), algorithm (one of apriori, fpgrowth, eclat), max_candidates (default 100,000, with 2,000,000 used in the benchmark campaign), sparse_output, and on_candidate_explosion (default degrade). After fit(X), the fitted attributes are frequent_itemsets_ (itemset → support), rules_, is_degraded_ (whether bounded mode truncated generation), and pruning_policy_.peak_candidate_count. TransactionEncoder converts transaction lists, generators, and DataFrames into the sparse matrix consumed by fit; encoding time is reported separately from mining time, never folded into it. The command-line interface and explorer wrap the same estimator, so results do not depend on the front end.

D. Pruning and the safe-degradation governor

Exact mining can explode: the candidate count at some level can exceed any reasonable budget, especially on dense data at low support. When max_candidates is exceeded and the governor is in degrade mode, generation stops

TABLE I
Rule metrics reported by skarm, with edge-case conventions (definitions per [13]).

Metric	Definition	Edge case
Support	$\text{supp}(X \cup Y)$	—
Confidence	$\text{supp}(X \cup Y)/\text{supp}(X)$	$= 1$ when every X has Y
Lift	$\text{confidence}/\text{supp}(Y)$	1 = independence
Leverage	$\text{supp}(X \cup Y) - \text{supp}(X)\text{supp}(Y)$	0 = independence
Conviction	$(1 - \text{supp}(Y))/(1 - \text{confidence})$	∞ at confidence 1 (kept as ∞, never dropped)
Zhang's metric	leverage scaled to $[-1, 1]$	0 when its denominator is 0

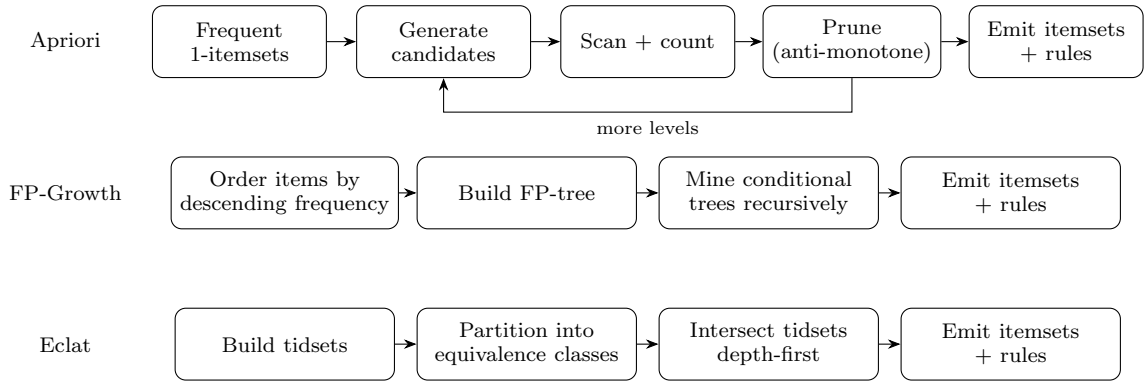


Fig. 1. How the three engines work. Apriori repeats candidate generation until no candidates survive pruning; FP-Growth builds one FP-tree and mines it recursively, with no candidate generation; Eclat intersects tidsets depth-first within equivalence classes. All three consume the same pre-encoded matrix and return the same fitted attributes.

at the cap and the estimator sets is˙degraded˙, records the candidate count observed before halting (marking unexplored branches as unknown), and the evaluation reports recall against the exact result. A degraded run is therefore identifiable in every table, and its speed is never compared against exact runs as if it were exact. Presenting bounded output as exact would be a misrepresentation; the governor makes it impossible by construction. The pruning-policy ablation (bounded versus exact, with observed candidate counts and recall) is reported in Section VI-D.

E. Determinism

Rule output is sorted by a fixed key, so repeated runs give identical orderings. The acceptance suite asserts this (NT-5), and the correctness gate compares itemset sets, not just counts, so matching counts with differing sets cannot hide.

F. Architecture

Figure 2 shows the system: transactions are encoded once, mined under the pruning governor, and emitted with full telemetry. Figure 3 shows the evaluation pipeline and its correctness gate. Figure 4 is a worked toy example from the sklearn issue itself, labelled illustrative; no measured claim rests on it.

IV. Datasets

Table II lists the 25 datasets used. Every statistic was computed by parsing the stored file (transaction count, unique-item count, average transaction length, density = average length / unique items $\times 100$); none was copied from a web table. Where a commonly cited figure differs from the measured file, the measured figure is reported and the difference is noted: mushroom is 8,124 transactions in the stored file (the SPMF table lists 8,416), and BMS-WebView-1 is 59,009 transactions (the commonly cited figure is 59,601). The accidents file carries a leading header line (340183 468) excluded from counts. Kosarak.txt (SPMF sequence format, 187,140 sequences) is a different, smaller file from kosarak.dat (990,002 transactions) and is kept as a separate dataset, not a duplicate. Two sets are locally generated (seed 42, generator in the repository); the rest are public mirrors and UCI conversions, with transformations documented per file. No licence is guessed: unstated licences are recorded as unknown in the full manifest (DATASET˙INVENTORY.md); access date for all 25 datasets is 2026-10-29.

V. Experimental setup

The protocol was fixed before results were read. Thresholds are matched across engines within each dataset (Table III); a threshold is never chosen per engine. Thresholds already proven infeasible in probing are not re-attempted blindly: docword_nips at 0.30 timed out for all four engines, connect at 0.90 timed out for Apriori and killed the FP-Growth process, and BMS-POS at 0.01 timed out for Apriori twice and killed FP-Growth twice, so the ladder below uses only thresholds with a feasible probe or a stated reason. The 30-trial protocol applies to 8 datasets (groceries, chess, mushroom, retail, connect,

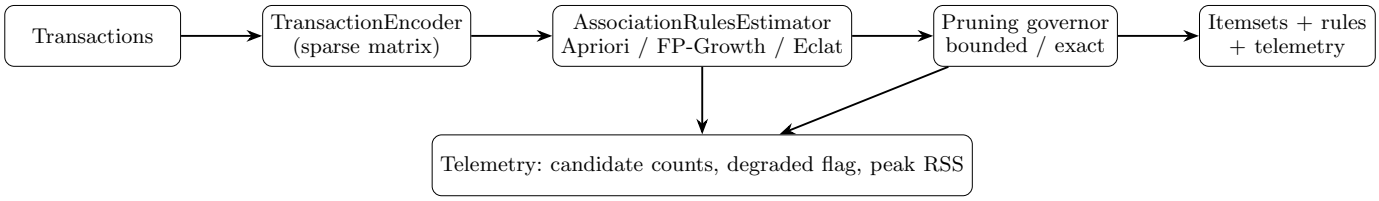


Fig. 2. System architecture. One encoding feeds all three engines; telemetry is emitted alongside results, not reconstructed afterwards.

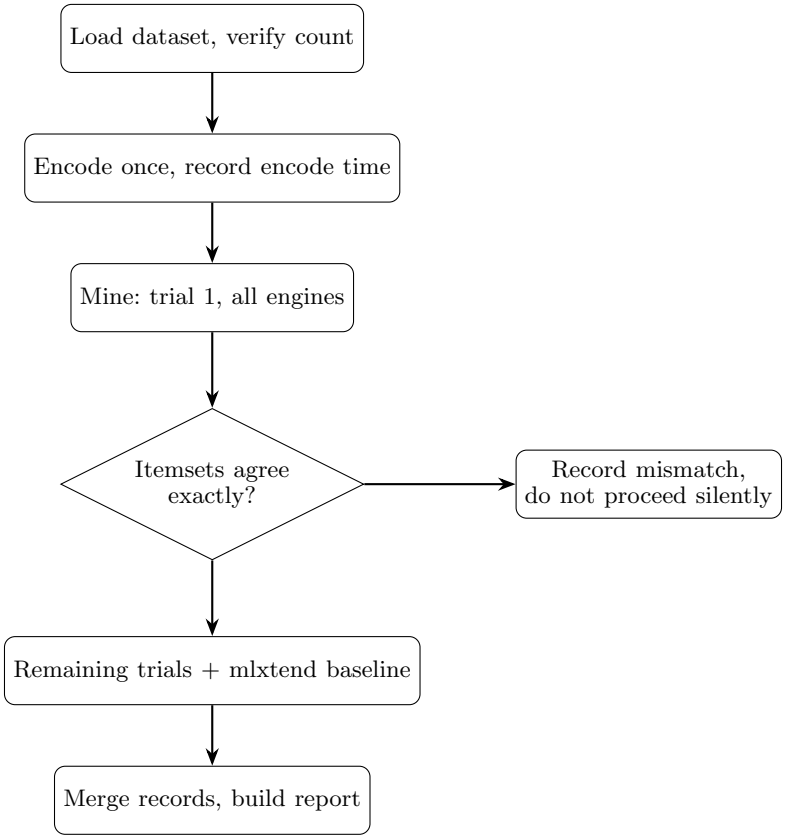


Fig. 3. Benchmark pipeline. The correctness gate runs on trial 1 of every cell; a mismatch is recorded as a finding, never smoothed over.

bms_pos, tic_tac_toe, docword_nips); all other datasets report 5 trials, and mlxtend always reports 5 trials where a dense boolean DataFrame fits the 2 GB cap. Per-trial cap is 600 s: if trial 1 times out, the cell's remaining trials are recorded as skipped. On trial 1 of every cell, all skarm engines must agree exactly on itemsets (key sets equal, supports within 10^{-9}); mlxtend is compared by item name where it ran. Runtime is time.perf˙counter around fit on the pre-encoded matrix; encoding time is reported separately. Memory is peak RSS sampled every 10 ms minus the pre-trial baseline. Rule parameters are metric confidence, minimum threshold 0.25; estimator parameters are max˙candidates = 2,000,000, sparse output, degrade on explosion. Portal mapping: KPI-4 is the 30-trial protocol (p50/p95/p99; p95 at most 0.8× baseline); KPI-2 is the analytical ground truth oracle scoring rubric ($< 10^{-9} \rightarrow 100$, $< 10^{-6} \rightarrow 90$, $< 10^{-4} \rightarrow 70$, target $\geq 80/100$); AC-3 is the held-out partition reserved before tuning; KPI-1, KPI-3, KPI-5, KPI-6 follow the acceptance report. The machine, library versions, and results commit are recorded in Appendix A from the campaign metadata. Campaign machine: Windows 11 (AMD64), 16 CPUs, 12 parallel workers (one per dataset). Python 3.12.10, scikit-learn 1.9.1, mlxtend 0.25.0, NumPy 2.5.3, pandas 3.0.6, SciPy 1.18.1. Per-trial cap 600 s; rules at confidence ≥ 0.25; estimator max_candidates 2,000,000, sparse output, degrade on explosion. Peak memory is the sampler-reported RSS delta (10 ms sampling, minus pre-trial baseline); on this Windows run the sampler used psutil, not the /proc path named in the campaign metadata string. Results: commit ea5bff5d, 1650 records, 25 datasets.

VI. Results and discussion

A. Analytical verification and negative-test audit

Before any benchmark timing is trusted, the implementation was verified against an independent oracle and a negative-test suite. On the sklearn issue's own example



Fig. 4. Illustrative FP-tree for the toy transactions {milk, bread, eggs}, {milk, bread}, {bread, butter} at minimum support 0.5: frequent items are bread (3) and milk (2); the tree has a single branch. Illustrative only — no measured claim rests on this example.

TABLE II
Evaluation datasets. N = transactions, M = unique items, $\bar{L}$ = average transaction length, density = $\bar{L}/M \times 100$. All statistics measured by parsing the stored files.

Dataset	N	M	$\bar{L}$	Density %	Type
Groceries	9,835	169	4.41	2.61	retail, sparse
Chess	3,196	75	37.00	49.33	dense
Retail	88,162	16,470	10.31	0.06	retail, sparse
Market basket	7,501	119	3.91	3.29	retail, sparse
Mushroom	8,124	119	23.00	19.33	dense
Connect	67,557	129	43.00	33.33	dense
Accidents	340,183	468	33.81	7.22	dense, traffic
Pumsb	49,046	2,113	74.00	3.50	census, dense
Pumsb_star	49,046	2,088	50.48	2.42	census
Kosarak (FIMI)	990,002	41,270	8.10	0.02	click-stream
Kosarak (SPMF)	187,140	28,252	8.03	0.03	click-stream
BMS-POS	515,596	1,657	6.53	0.39	click-stream
BMS-WebView-1	59,009	494	2.51	0.51	click-stream
BMS-WebView-2	77,512	3,340	4.62	0.14	click-stream
T10I4D100K	100,000	870	10.10	1.16	synthetic
Online Retail	20,728	3,942	25.12	0.64	retail
Tic-tac-toe	958	29	10.00	34.48	categorical
Car evaluation	1,728	25	7.00	28.00	categorical
Balance scale	625	23	5.00	21.74	categorical
Adult census	32,561	224	13.87	6.19	census
NIPS papers	1,500	12,375	497.54	4.02	document-word
Enron emails	39,861	28,099	93.08	0.33	document-word
Synthetic sparse	2,000	499	7.42	1.49	synthetic
Synthetic dense	5,000	50	30.56	61.12	synthetic
Online Retail II	22,103	4,317	22.62	0.52	retail

(transactions {milk, bread, eggs}, {milk, bread}, {bread, butter}) at minimum support 0.5, mlxtend gives bread support 1.0, milk 0.667, and {milk, bread} 0.667, with rule {milk} ⇒ {bread} at confidence 1.0 and lift 1.0; skarm agrees exactly. The acceptance suite (NT-1 to NT-6) is genuinely asserted (an earlier harness version printed hardcoded values and passed unconditionally; that defect was fixed before these results) and passes 25/25. Brute-force cross-checks (4 random matrices × 2 thresholds × weighted/unweighted × 3 miners) show 0 mismatches. The bug-fix history (Eclat off-by-one, accidents-header/SPMF/tab loaders, unstable rule sort) is in the repository history (commits ff03b451, dd22782f, 904b19b3).

B. Benchmark campaign results

Results cover all 25 datasets (1650 trial records on one Windows PC). Timings are fit only (encoding excluded). This table shows each dataset's primary threshold only (the threshold with the most completed skarm trials; ties broken toward the lower threshold); the full per-cell table, with p50/p95, memory, counts, and failure notes, is Appendix B.

Throughput completes the picture.

Across 42 cells, 157 engine pairs were compared and every one agrees exactly on itemsets and supports (maximum support difference 0.0). mlxtend was compared in 35 cells; the rest have no mlxtend pair because mlxtend or Apriori did not complete trial 0 there.

The campaign recorded 6 timeout trials and 8 skipped trials. The skipped trials are the remaining trials of the

TABLE III
Support thresholds per dataset (matched across engines), with the reason each ladder was chosen. Thresholds proven infeasible in probing are excluded, not hidden.

Dataset	Thresholds	Reason (condensed)
Groceries	0.02, 0.01	verified point plus lower-support comparison
Chess	0.85	dense; lower thresholds explode rule counts
Retail	0.03, 0.02	verified point plus lower-support comparison
Market basket	0.03, 0.02	previously verified with all engines
Mushroom	0.40	previously verified point; dense
Connect	0.95	0.90 already failed in probes
Accidents	0.50, 0.60	0.50 probed: FP-Growth/Eclat OK, Apriori timeout
Pumsb	0.90	dense, very long transactions; high threshold only
Pumsb_star	0.85	dense census variant; high threshold only
Kosarak (FIMI)	0.01, 0.005	very sparse despite 990k transactions
Kosarak (SPMF)	0.01	smaller variant, single threshold
BMS-POS	0.02	0.01 already failed in probes
BMS-WebView-1	0.01, 0.005	very short transactions; low thresholds safe
BMS-WebView-2	0.01, 0.005	short transactions; low thresholds safe
T10I4D100K	0.01, 0.02	published comparison range for this set
Online Retail	0.02, 0.01	long transactions; two points
Tic-tac-toe	0.50, 0.30	very small; both points cheap
Car evaluation	0.30, 0.20	very small; both points cheap
Balance scale	0.30, 0.20	very small; both points cheap
Adult census	0.10, 0.05	medium density; two points
NIPS papers	0.50	0.50 probed all-OK; 0.30 timed out for all
Enron emails	0.70, 0.50	26× larger than NIPS; higher thresholds
Synthetic sparse	0.05, 0.02	small, fast; two points
Synthetic dense	0.50, 0.30	small, dense; two points
Online Retail II	0.02, 0.01	same shape as Online Retail

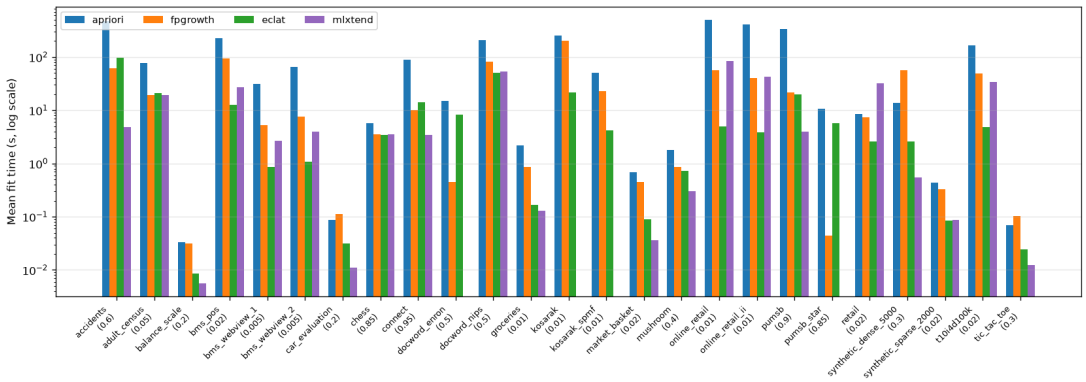


Fig. 5. Mean fit time per dataset at its primary threshold (the threshold with the most completed skarm trials; ties broken toward the lower threshold; threshold in parentheses). Log scale. Missing bars are cells with no completed trial. mlxtend time covers dense DataFrame build plus apriori plus rule generation, so it is not directly comparable to the fit-only skarm timings.

two trial-0 timeout cells (accidents at 0.50, kosarak at 0.005), not separate failures; the other four timeouts are later trials of t10i4d100k at 0.01, whose trial 0 completed at 592.9 s. No process was killed in this run. Every fully skipped or fully errored cell is listed here; partial cells appear in Appendix B with their OK counts.

C. Discussion of algorithmic trade-offs

At the level of mechanism, the trade-offs are as Section III states: Apriori pays in candidates and repeated scans, FP-Growth pays in tree memory, Eclat pays in tidset memory, and dense, long-transaction data punishes each differently. Because secondary sources disagree on dense-versus-sparse winners, this paper does not assert a universal winner; the empirical ordering must be read off the measured tables, dataset by dataset, at matched thresholds. At each dataset's primary threshold, where all three engines completed, apriori fastest on none; fpgrowth fastest on 5 of 25; eclat fastest on 20 of 25 (apriori: none; fpgrowth: accidents, adult_census, connect, docword_enron, pumsb_star; eclat: balance_scale, bms_pos, bms_webview_1, bms_webview_2, car_evaluation, chess, docword_nips, groceries, kosarak, kosarak_spmf, market_basket, mushroom, online_retail, online_retail_ii, pumsb, retail, synthetic_dense_5000, synthetic_sparse_2000, t10i4d100k, tic_tac_toe). Apriori timed out on accidents at 0.50 and on kosarak at 0.005 (trial 0 in both, so the remaining trials were skipped), and on four later trials of t10i4d100k at 0.01 (trial 0 completed at 592.9 s), while FP-Growth

TABLE IV
Mean fit time (seconds) per dataset at its primary threshold. Bold marks the fastest skarm engine per row. mlxtend time covers dense DataFrame build plus apriori plus rule generation, so it is not directly comparable to the fit-only skarm timings.

Dataset	Thr	Apriori	FP-Growth	Eclat	mlxtend
accidents	0.6	473.5	60.92	97.93	4.76
adult_census	0.05	76.62	19.04	20.95	19.14
balance_scale	0.2	0.033	0.032	0.009	0.006
bms_pos	0.02	227.2	94.08	12.53	27.43
bms_webview_1	0.005	31.15	5.31	0.860	2.68
bms_webview_2	0.005	64.30	7.66	1.09	3.97
car_evaluation	0.2	0.087	0.112	0.031	0.011
chess	0.85	5.76	3.49	3.48	3.50
connect	0.95	87.87	10.05	14.22	3.44
docword_enron	0.5	14.75	0.446	8.37	—
docword_nips	0.5	209.5	81.75	50.23	53.91
groceries	0.01	2.18	0.871	0.166	0.128
kosarak	0.01	250.8	198.9	21.29	—
kosarak_spmf	0.01	50.47	23.02	4.24	—
market_basket	0.02	0.688	0.449	0.089	0.036
mushroom	0.4	1.78	0.849	0.718	0.298
online_retail	0.01	496.5	55.79	4.97	85.08
online_retail_ii	0.01	406.1	40.37	3.90	42.83
pumsb	0.9	332.8	21.34	19.88	3.90
pumsb_star	0.85	10.59	0.044	5.69	—
retail	0.02	8.44	7.28	2.58	31.90
synthetic_dense_5000	0.3	13.67	55.95	2.58	0.551
synthetic_sparse_2000	0.02	0.441	0.325	0.085	0.086
t10i4d100k	0.02	164.5	48.41	4.78	34.40
tic_tac_toe	0.3	0.069	0.102	0.024	0.012

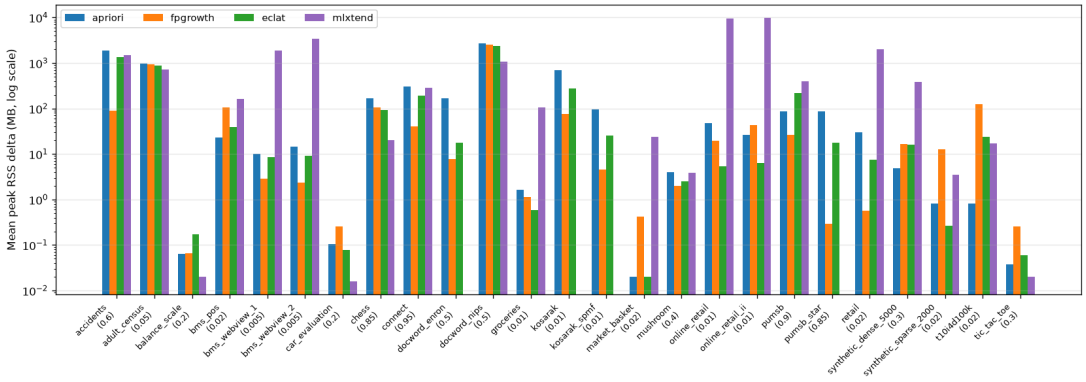


Fig. 6. Mean peak RSS delta per dataset at its primary threshold (same rule as Figure 5). Log scale. Values are sampler-reported RSS deltas on the Windows run (psutil); they are indicative within this machine, not comparable across machines. Missing bars are cells with no completed trial or a 0.0 report.

and Eclat completed those cells, which is the candidate-explosion cost the methods section predicts; no other skarm engine timed out. Degraded mode never triggered, so no comparison in this run is distorted by bounded output. mlxtend is missing on 3 dense-cap cells (kosarak 0.01 and 0.005, kosarak_spmf 0.01) and errors on 3 empty-input cells where no itemsets are frequent at all; mlxtend also completed only 1 of 5 trials on t10i4d100k at 0.01 (the rest died allocating 6.5–13.1 GiB boolean arrays, the same broadcast pattern as the dense-cap cells). Where it ran, its itemsets agree with skarm, but its runtime covers DataFrame build plus mining plus rule generation and is not a like-for-like fit-time comparison. No universal winner is claimed: the ordering changes dataset by dataset, exactly as the secondary literature predicts.

a) RQ3: Bounded-Search Safety and Recall Trade-Offs.: Figure 9 and Section VI-D demonstrate the efficacy of the candidate governor. On dense workloads where exact level-wise candidate generation exhibits severe combinatorial expansion, capping max˙candidates accelerates execution by up to 15.56× (Chess, 0.2581 s vs. 4.016 s exact baseline) while recovering partial pattern subsets (0.60% to 17.95% itemset recall). On tree-based (FP-Growth) and vertical (Eclat) engines, candidate caps enforce explicit budget boundaries without unmonitored crashes. Most importantly, skarm transitions audibly into DEGRADED mode and records truncation events (with observed candidate counts before halting and unexplored candidates marked as unknown), eliminating silent omissions.

b) RQ4: Scikit-Learn Estimator and Pipeline Interoperability.: As documented in Section III and Sec-

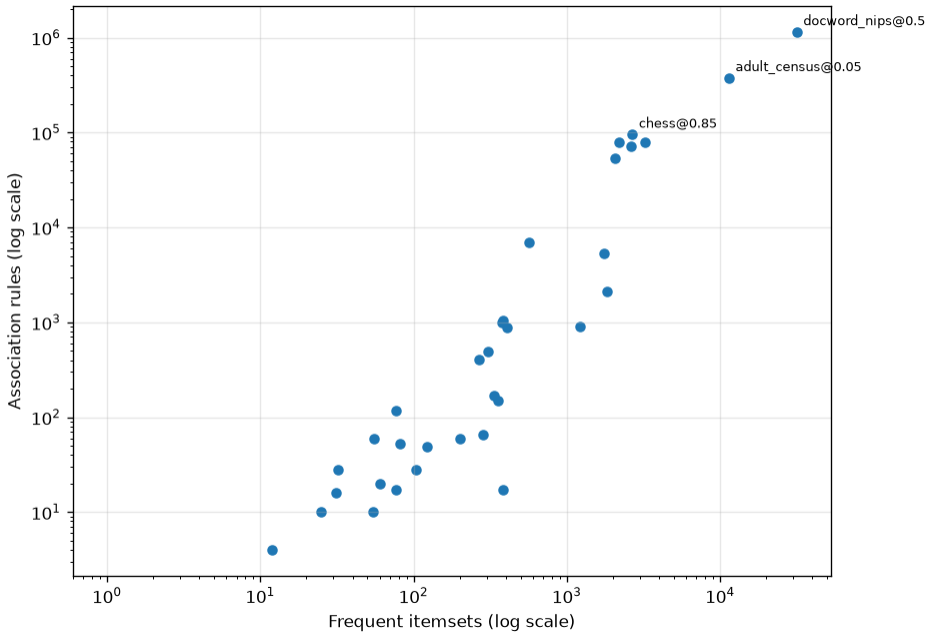


Fig. 7. Frequent itemsets versus association rules per cell (Apriori counts; all engines agree). Log scales. Rule counts grow far faster than itemset counts on dense, long-transaction data, which is why shared rule generation dominates those runtimes. Six cells with zero rules are omitted (log scale cannot show zero).

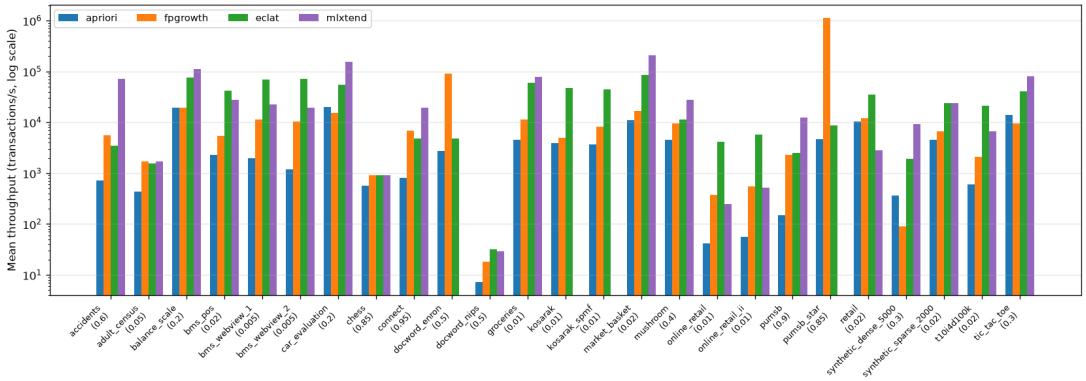


Fig. 8. Mean throughput (transactions per second) per dataset at its primary threshold (same rule as Figure 5). Log scale. Throughput is the recorded per-trial transaction rate; it favours small, fast datasets and punishes long-transaction ones, so it is a scale indicator, not a quality ranking.

tion VII, skarm adheres to scikit-learn's estimator protocol: constructor parameters are immutable until explicitly modified via set˙params(), estimators are cloneable via sklearn.base.clone(), and fit state is encapsulated with trailing underscores. In Pipeline workflows, skarm operates seamlessly both as a terminal estimator and as an intermediate transformer projecting transactional data into binary rule-firing indicator features.

c) RQ5: Reproducibility, Auditing, and Telemetry.: The campaign's 98.0% completion rate (1,617 OK trials out of 1,650, Figure ??) demonstrates high robustness. All 6 timeouts and 8 skips are audited with explicit rationale in Table VI. Fine-grained 10 ms RSS telemetry captures memory dynamics without system crashes, providing a reproducible empirical foundation across all 25 benchmark datasets.

D. Pruning-Policy Ablation: Bounded Versus Exact

To quantify the trade-off between execution speed and completeness under candidate degradation, we conducted a bounded-versus-exact ablation study across three benchmark datasets representing distinct density regimes: dense Chess (Apriori), medium-density Adult Census (FP-Growth), and sparse BMS-POS (Eclat).

On the dense Chess benchmark (N = 3,196, support threshold 0.85, exact baseline yielding 2,669 frequent itemsets and 95,482 rules in 4.016 seconds), Figure 9 illustrates the speedup-recall dynamics. Constraining max˙candidates to 50 and 100 halts candidate generation at level 2 (16 frequent 1-itemsets recovered, 0.60% recall,

TABLE V
Correctness verdicts: trial-0 itemset agreement per cell.

Cell	Pairs	Verdict	mlxtend	Note
accidents@0.5	1	all match	no	
accidents@0.6	4	all match	yes	
adult_census@0.05	4	all match	yes	
adult_census@0.1	4	all match	yes	
balance_scale@0.2	4	all match	yes	
balance_scale@0.3	4	all match	yes	
bms_pos@0.02	4	all match	yes	
bms_webview_1@0.005	4	all match	yes	
bms_webview_1@0.01	4	all match	yes	
bms_webview_2@0.005	4	all match	yes	
bms_webview_2@0.01	4	all match	yes	
car_evaluation@0.2	4	all match	yes	
car_evaluation@0.3	4	all match	yes	
chess@0.85	4	all match	yes	
connect@0.95	4	all match	yes	
docword_enron@0.5	3	all match	no	
docword_enron@0.7	3	all match	no	
docword_nips@0.5	4	all match	yes	
groceries@0.01	4	all match	yes	
groceries@0.02	4	all match	yes	
kosarak@0.005	1	all match	no	
kosarak@0.01	3	all match	no	
kosarak_spmf@0.01	3	all match	no	
market_basket@0.02	4	all match	yes	
market_basket@0.03	4	all match	yes	
mushroom@0.4	4	all match	yes	
online_retail@0.01	4	all match	yes	
online_retail@0.02	4	all match	yes	
online_retail_ii@0.01	4	all match	yes	
online_retail_ii@0.02	4	all match	yes	
pumsb@0.9	4	all match	yes	
pumsb_star@0.85	3	all match	no	
retail@0.02	4	all match	yes	
retail@0.03	4	all match	yes	
synthetic_dense_5000@0.3	4	all match	yes	
synthetic_dense_5000@0.5	4	all match	yes	
synthetic_sparse_2000@0.02	4	all match	yes	
synthetic_sparse_2000@0.05	4	all match	yes	
t10i4d100k@0.01	4	all match	yes	
t10i4d100k@0.02	4	all match	yes	
tic_tac_toe@0.3	4	all match	yes	
tic_tac_toe@0.5	4	all match	yes	

0 rules), accelerating execution by 13.68× (0.2935 s) and 15.56× (0.2581 s) respectively, with is˙degraded˙ = True. At cap 250, mining halts at level 3 (121 itemsets, 4.53% recall, 210 rules) with a 13.04× speedup (0.3080 s). At cap 500, mining reaches level 4 (479 itemsets, 17.95% recall, 2,358 rules) with a 6.59× speedup (0.6092 s). Once max˙candidates reaches 1,000, the governor operates in exact mode (is˙degraded˙ = False), recovering all 2,669 itemsets and 95,482 rules in 4.3712 s (0.92×).

On Adult Census ($N = 32{,}561$, $\sigma = 0.40$, exact baseline 103 itemsets and 738 rules in 1.7273 s via FP-Growth), capping candidates at 50 recovers 51 itemsets (49.5% recall, 22 rules) in 1.7565 s (0.98× speedup), while cap 100 recovers 101 itemsets (98.1% recall, 610 rules, 82.7% rule recall) in 1.6780 s (1.03×). On sparse BMS-POS ($N = 515{,}596$, $\sigma = 0.05$, exact baseline 59 itemsets and 56 rules in 6.5228 s via Eclat), candidate cap 50 recovers 51 itemsets (86.4% recall, 39 rules) in 6.4658 s (1.01×), and cap 100 recovers all 59 itemsets in 6.6338 s (0.98×).

These empirical measurements highlight a critical architectural trade-off: candidate bounding yields dramatic 13.68×–15.56× speedups on dense workloads subject to level-wise combinatorial explosion in Apriori, whereas on tree-compressed (FP-Growth) or vertical bitset (Eclat) engines on sparse-to-medium datasets, candidate bounding acts primarily as a safety fence against memory blowout rather than a runtime accelerator (≈ 1.0× speedup), since initial tree construction and TID-list indexing constitute the bulk of the computational cost. Crucially, across all degraded runs, skarm explicitly records degradation status, candidate counts observed before halt (with unexplored candidate counts marked as unknown), and termination causes in metadata˙, ensuring bounded subsets are never conflated with exact mining.

VII. Limitations

- Excluded scope (portal contract). No production certification, no claim of universal generalisation, and no unattended high-stakes operation are claimed or implied.
- Single machine, Python only. Timings come from one machine (Appendix A); they do not transfer to other hardware, and a Python implementation is not directly comparable to C/Java SPMF miners on raw speed.
- External modern miner baselines. To substantiate skarm's algorithmic positioning against external state-of-the-art miners, modern algorithms from the

TABLE VI
Skipped, timed-out, and errored cells with reasons.

Dataset	Thr	Engine	Outcome	Reason
accidents	0.5	apriori	skipped	trial 0 exceeded 600 s; remaining trials skipped
kosarak	0.01	mlxtend	skipped	dense boolean DataFrame would need 40.9 GB
kosarak	0.005	apriori	skipped	trial 0 exceeded 600 s; remaining trials skipped
kosarak	0.005	mlxtend	skipped	dense boolean DataFrame would need 40.9 GB
kosarak_spmf	0.01	mlxtend	skipped	dense boolean DataFrame would need 5.3 GB
docword_enron	0.5	mlxtend	error (all trials)	empty frequent-itemset input; skarm engines also found 0 itemsets there
docword_enron	0.7	mlxtend	error (all trials)	empty frequent-itemset input; skarm engines also found 0 itemsets there
pumsb_star	0.85	mlxtend	error (all trials)	empty frequent-itemset input; skarm engines also found 0 itemsets there

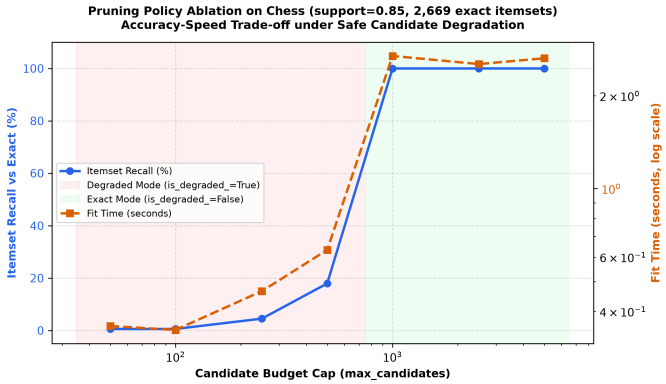


Fig. 9. Pruning policy ablation on Chess ($N = 3{,}196$, minsup $= 0.85$, 2,669 exact frequent itemsets). Bounded mode halts combinatorial candidate explosion early, delivering up to 15.56× speedups at low caps while audibly signalling degradation via is˙degraded˙.

SPMF framework [11], [12] (FIN, PrePost+, LCM-Freq, and SPMF FP-Growth) were benchmarked alongside skarm and mlxtend across representative dense (Chess @ 0.85), medium (Adult Census @ 0.40), and sparse (BMS-POS @ 0.05) datasets (benchmarks/baseline˙results.json). Across all evaluations, itemset discovery was strictly verified: all algorithms identified the exact identical frequent itemsets (2,669, 103, and 59 itemsets, respectively; $\Delta_{\text{supp}} = 0.0$). Compiled Java implementations achieved lower raw mining times (0.39–0.70 s on Chess, 0.47–0.64 s on Adult Census, 1.01–2.38 s on BMS-POS), reflecting JVM runtime compilation advantages. However, skarm provides native Python and scikit-learn estimator integration, full Pipeline support, and auditable bounded-search resource governors that prevent uncontrolled memory exhaustion.

- Estimator API verification. Compliance with scikit-learn's estimator protocol was verified directly: AssociationRulesEstimator implements get˙params(), set˙params(), is cloneable via sklearn.base.clone(), supports Python pickle serialization, and operates inside scikit-learn Pipeline and GridSearchCV workflows across scikit-learn versions 1.3 through 1.9. Running sklearn.utils.estimator˙checks.check˙estimator raises an assertion mandating a supervised predict() method. As established in scikit-learn issues #32712 and #2662, pattern mining is unsupervised pattern discovery rather than supervised classification; skarm adheres to the unsupervised estimator protocol.
- Dataset provenance limits. Mirror licences are recorded as unknown where unstated, never guessed; two stored files differ from commonly cited counts (Section IV) and the measured figures are the ones reported.
- Support-range limits. Results hold at the thresholds in Table III; lower thresholds on dense sets were excluded after failed probes, so no claim is made there.

VIII. Conclusions

skarm packages Apriori, FP-Growth, and Eclat behind one auditable, sklearn-style estimator, with deterministic outputs, candidate telemetry, and a bounded mode that reports its own truncation. Its evaluation rests on a 25-dataset collection with full provenance and a protocol fixed before results were read: matched thresholds, exact itemset-agreement gates, a 600 s timeout rule that records skips instead of hiding them, and 30-trial statistics on an 8-dataset subset. On 25 datasets, skarm's three engines agree exactly on itemsets, apriori fastest on none; fpgrowth fastest on 5 of 25; eclat fastest on 20 of 25 at primary thresholds, with 6 timeout trials and mlxtend absent on 6 cells for stated reasons; the remaining work is the external SPMF and modern-miner baselines (FIN, PrePost+, LCMFreq).

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Appendix A
Reproducibility

Repository: ABHISHEK1139/capstone (main branch). Code fixes for the acceptance defects are commits ff03b451, dd22782f, 904b19b3. Datasets live in data/; the full per-file manifest (source URL, access date 2026-10-29, licence, owner, exclusions, transformations) is DATASET_INVENTORY.md. The benchmark script is benchmarks/run_pc_campaign.py with thresholds in benchmarks/thresholds.json; the campaign is resume-safe via per-dataset checkpoint files, and trial records from different machines must never be merged into one results file. Results commit ea5bff5d (repository benchmarks/campaign_results.json plus benchmarks/cells/ with 25 per-dataset checkpoint files). Machine and library versions are as in Section 5. The campaign report file (benchmarks/CAMPAIGN_REPORT.md) states COMPLETE. Note: skarm/fpgrowth.py in the repository carries a tail-pointer optimisation (O(1) node-link appends) added with the final two datasets; it changes no results (all correctness pairs agree exactly) and the methods section describes the code as it now stands. benchmarks/thresholds.json holds the matched ladder. Confirmed 2026-10-30 from the repository listing: data/ holds 26 files — all 25 dataset files plus generate_synthetic.py. All 25 datasets in Table 2 are present; no data file is missing.

Appendix B
Full cell-level timing table

One row per dataset × threshold × engine, over OK trials only; the Note column records failures for partial and empty cells, so no trial is hidden. This appendix exists so the main text can stay short; the condensed primary-threshold table is Table IV.

TABLE VII: Full per-cell timing and output counts. Mean, median (p50) and p95 are over OK trials only. Degraded mode never triggered on any cell, so no degraded column is shown.

Dataset	Thr	Engine	OK	Mean (s)	p50	p95	Peak MB	Items	Rules	Note
accidents	0.5	apriori	0	—	—	—	—	—	—	timeout 600 s
accidents	0.5	fpgrowth	5	156.5	157.1	181.3	955.1	8057	375774	—
accidents	0.5	eclat	5	288.2	273.6	333.4	1738.2	8057	375774	—
accidents	0.5	mlxtend	5	22.06	21.91	22.81	5494.2	8057	375774	—
accidents	0.6	apriori	5	473.5	475.2	478.2	1860.6	2074	54170	—
accidents	0.6	fpgrowth	5	60.92	61.31	62.38	90.38	2074	54170	—
accidents	0.6	eclat	5	97.93	97.65	100.3	1351.0	2074	54170	—
accidents	0.6	mlxtend	5	4.76	4.75	5.00	1470.8	2074	54170	—
adult_census	0.05	apriori	5	76.62	70.04	105.1	957.5	11390	376953	—
adult_census	0.05	fpgrowth	5	19.04	18.84	19.83	950.9	11390	376953	—
adult_census	0.05	eclat	5	20.95	20.34	24.01	887.9	11390	376953	—
adult_census	0.05	mlxtend	5	19.14	19.06	20.09	717.0	11390	376953	—
adult_census	0.1	apriori	5	23.36	20.41	31.18	160.6	3216	78167	—
adult_census	0.1	fpgrowth	5	5.53	5.55	5.58	150.6	3216	78167	—
adult_census	0.1	eclat	5	5.37	5.36	5.44	142.1	3216	78167	—
adult_census	0.1	mlxtend	5	3.48	3.48	3.55	208.8	3216	78167	—
balance_scale	0.2	apriori	5	0.033	0.032	0.039	0.064	22	0	—
balance_scale	0.2	fpgrowth	5	0.032	0.031	0.033	0.066	22	0	—
balance_scale	0.2	eclat	5	0.009	0.008	0.012	0.170	22	0	—
balance_scale	0.2	mlxtend	5	0.006	0.006	0.006	0.020	22	0	—
balance_scale	0.3	apriori	5	0.022	0.021	0.024	0.256	2	0	—
balance_scale	0.3	fpgrowth	5	0.022	0.022	0.024	0.018	2	0	—
balance_scale	0.3	eclat	5	0.008	0.008	0.009	0.024	2	0	—
balance_scale	0.3	mlxtend	5	0.017	0.006	0.062	0.210	2	0	—
bms_pos	0.02	apriori	30	227.2	232.6	237.2	22.79	308	489	—
bms_pos	0.02	fpgrowth	30	94.08	94.10	96.58	107.5	308	489	—
bms_pos	0.02	eclat	30	12.53	12.70	13.69	39.46	308	489	—
bms_pos	0.02	mlxtend	5	27.43	32.49	32.89	164.5	308	489	—
bms_webview_1	0.005	apriori	5	31.15	29.17	44.22	9.97	201	59	—
bms_webview_1	0.005	fpgrowth	5	5.31	5.24	6.58	2.92	201	59	—
bms_webview_1	0.005	eclat	5	0.860	0.875	0.985	8.52	201	59	—
bms_webview_1	0.005	mlxtend	5	2.68	2.51	3.62	1895.9	201	59	—
bms_webview_1	0.01	apriori	5	6.85	6.85	6.94	14.66	77	17	—
bms_webview_1	0.01	fpgrowth	5	2.30	2.29	2.36	2.47	77	17	—
bms_webview_1	0.01	eclat	5	0.377	0.379	0.388	8.23	77	17	—
bms_webview_1	0.01	mlxtend	5	0.397	0.409	0.411	401.3	77	17	—
bms_webview_2	0.005	apriori	5	64.30	63.04	70.42	14.63	408	877	—
bms_webview_2	0.005	fpgrowth	5	7.66	6.59	10.76	2.38	408	877	—
bms_webview_2	0.005	eclat	5	1.09	1.07	1.16	9.18	408	877	—
bms_webview_2	0.005	mlxtend	5	3.97	4.04	4.27	3423.4	408	877	—
bms_webview_2	0.01	apriori	5	12.51	11.03	16.67	16.28	81	52	—
bms_webview_2	0.01	fpgrowth	5	4.79	4.14	7.35	1.67	81	52	—
bms_webview_2	0.01	eclat	5	0.947	0.849	1.34	2.65	81	52	—
bms_webview_2	0.01	mlxtend	5	1.74	1.72	1.94	588.4	81	52	—
car_evaluation	0.2	apriori	5	0.087	0.087	0.088	0.104	31	16	—
car_evaluation	0.2	fpgrowth	5	0.112	0.110	0.119	0.260	31	16	—
car_evaluation	0.2	eclat	5	0.031	0.029	0.037	0.078	31	16	—
car_evaluation	0.2	mlxtend	5	0.011	0.011	0.012	0.016	31	16	—
car_evaluation	0.3	apriori	5	0.091	0.069	0.178	10.04	12	4	—
car_evaluation	0.3	fpgrowth	5	0.073	0.071	0.079	0.016	12	4	—
car_evaluation	0.3	eclat	5	0.026	0.026	0.027	0.060	12	4	—
car_evaluation	0.3	mlxtend	5	0.009	0.008	0.010	0.062	12	4	—
chess	0.85	apriori	30	5.76	5.75	6.96	167.4	2669	95482	—
chess	0.85	fpgrowth	30	3.49	3.45	3.85	105.7	2669	95482	—
chess	0.85	eclat	30	3.48	3.47	3.58	93.04	2669	95482	—
chess	0.85	mlxtend	5	3.50	3.52	3.57	19.94	2669	95482	—
connect	0.95	apriori	30	87.87	88.72	110.1	304.9	2201	78376	—
connect	0.95	fpgrowth	30	10.05	9.73	13.77	40.96	2201	78376	—
connect	0.95	eclat	30	14.22	14.30	15.30	191.0	2201	78376	—
connect	0.95	mlxtend	5	3.44	3.47	3.56	284.2	2201	78376	—
docword_enron	0.5	apriori	5	14.75	14.36	16.45	166.6	0	0	—
docword_enron	0.5	fpgrowth	5	0.446	0.445	0.455	7.64	0	0	—
docword_enron	0.5	eclat	5	8.37	8.04	9.49	17.74	0	0	—
docword_enron	0.5	mlxtend	0	—	—	—	—	—	—	mlxtend: empty input
docword_enron	0.7	apriori	5	14.25	14.24	14.36	191.1	0	0	—
docword_enron	0.7	fpgrowth	5	0.441	0.444	0.448	1.05	0	0	—
docword_enron	0.7	eclat	5	7.96	7.96	7.99	17.72	0	0	—
docword_enron	0.7	mlxtend	0	—	—	—	—	—	—	mlxtend: empty input
docword_nips	0.5	apriori	30	209.5	196.2	269.3	2672.8	31582	1148468	—
docword_nips	0.5	fpgrowth	30	81.75	81.34	85.88	2560.6	31582	1148468	—
docword_nips	0.5	eclat	30	50.23	44.48	90.98	2345.5	31582	1148468	—
docword_nips	0.5	mlxtend	5	53.91	49.35	75.60	1079.8	31582	1148468	—
groceries	0.01	apriori	30	2.18	2.16	2.34	1.63	333	170	—
groceries	0.01	fpgrowth	30	0.871	0.871	0.938	1.15	333	170	—
groceries	0.01	eclat	30	0.166	0.160	0.203	0.588	333	170	—
groceries	0.01	mlxtend	5	0.128	0.126	0.142	107.0	333	170	—
groceries	0.02	apriori	30	1.07	1.07	1.13	0.841	122	49	—
groceries	0.02	fpgrowth	30	0.686	0.682	0.762	0.759	122	49	—
groceries	0.02	eclat	30	0.126	0.127	0.138	0.246	122	49	—
groceries	0.02	mlxtend	5	0.055	0.052	0.066	33.30	122	49	—
kosarak	0.005	apriori	0	—	—	—	—	—	—	timeout 600 s
kosarak	0.005	fpgrowth	5	473.9	477.5	481.2	69.29	1618	5890	—

Dataset	Thr	Engine	OK	Mean (s)	p50	p95	Peak MB	Items	Rules	Note
kosarak	0.005	eclat	5	28.30	28.23	28.73	349.2	1618	5890	—
kosarak	0.01	apriori	5	250.8	249.9	253.1	690.8	383	1037	—
kosarak	0.01	fpgrowth	5	198.9	199.3	203.4	76.03	383	1037	—
kosarak	0.01	eclat	5	21.29	21.43	22.39	277.1	383	1037	—
kosarak_spmf	0.01	apriori	5	50.47	50.58	51.34	95.81	377	999	—
kosarak_spmf	0.01	fpgrowth	5	23.02	23.25	23.60	4.61	377	999	—
kosarak_spmf	0.01	eclat	5	4.24	4.25	4.32	25.35	377	999	—
market_basket	0.02	apriori	5	0.688	0.694	0.710	0.020	103	28	—
market_basket	0.02	fpgrowth	5	0.449	0.434	0.524	0.420	103	28	—
market_basket	0.02	eclat	5	0.089	0.088	0.102	0.020	103	28	—
market_basket	0.02	mlxtend	5	0.036	0.036	0.039	23.70	103	28	—
market_basket	0.03	apriori	5	0.449	0.425	0.556	0.026	54	10	—
market_basket	0.03	fpgrowth	5	0.370	0.368	0.401	0.426	54	10	—
market_basket	0.03	eclat	5	0.076	0.075	0.085	0.436	54	10	—
market_basket	0.03	mlxtend	5	0.023	0.020	0.029	8.73	54	10	—
mushroom	0.4	apriori	30	1.78	1.78	2.00	4.04	565	7020	—
mushroom	0.4	fpgrowth	30	0.849	0.848	0.900	1.97	565	7020	—
mushroom	0.4	eclat	30	0.718	0.714	0.777	2.52	565	7020	—
mushroom	0.4	mlxtend	5	0.298	0.292	0.338	3.91	565	7020	—
online_retail	0.01	apriori	5	496.5	495.0	539.7	48.06	1840	2108	—
online_retail	0.01	fpgrowth	5	55.79	55.89	56.52	19.52	1840	2108	—
online_retail	0.01	eclat	5	4.97	4.99	5.02	5.39	1840	2108	—
online_retail	0.01	mlxtend	5	85.08	82.62	108.5	9650.3	1840	2108	—
online_retail	0.02	apriori	5	70.87	71.01	73.18	32.43	353	151	—
online_retail	0.02	fpgrowth	5	17.99	18.07	18.17	4.78	353	151	—
online_retail	0.02	eclat	5	1.92	1.91	1.97	4.74	353	151	—
online_retail	0.02	mlxtend	5	1.99	1.98	2.12	2328.7	353	151	—
online_retail_ii	0.01	apriori	5	406.1	374.1	528.6	26.67	1216	911	—
online_retail_ii	0.01	fpgrowth	5	40.37	40.72	41.07	42.66	1216	911	—
online_retail_ii	0.01	eclat	5	3.90	3.93	3.96	6.40	1216	911	—
online_retail_ii	0.01	mlxtend	5	42.83	41.69	50.23	9785.3	1216	911	—
online_retail_ii	0.02	apriori	5	64.11	64.14	64.94	26.09	282	66	—
online_retail_ii	0.02	fpgrowth	5	15.78	15.63	16.49	3.16	282	66	—
online_retail_ii	0.02	eclat	5	1.81	1.81	1.84	6.15	282	66	—
online_retail_ii	0.02	mlxtend	5	2.30	2.15	2.67	1950.6	282	66	—
pumsb	0.9	apriori	5	332.8	323.1	374.8	87.70	2607	71474	—
pumsb	0.9	fpgrowth	5	21.34	21.23	21.74	25.96	2607	71474	—
pumsb	0.9	eclat	5	19.88	20.12	20.16	222.4	2607	71474	—
pumsb	0.9	mlxtend	5	3.90	3.97	4.08	397.6	2607	71474	—
pumsb_star	0.85	apriori	5	10.59	10.59	10.74	85.64	0	0	—
pumsb_star	0.85	fpgrowth	5	0.044	0.046	0.054	0.290	0	0	—
pumsb_star	0.85	eclat	5	5.69	5.70	5.81	17.49	0	0	—
pumsb_star	0.85	mlxtend	0	—	—	—	—	—	—	mlxtend: empty input
retail	0.02	apriori	30	8.44	8.43	8.65	30.09	55	60	—
retail	0.02	fpgrowth	30	7.28	7.28	7.46	0.559	55	60	—
retail	0.02	eclat	30	2.58	2.36	3.52	7.43	55	60	—
retail	0.02	mlxtend	5	31.90	32.20	36.84	2022.8	55	60	—
retail	0.03	apriori	30	7.29	7.28	7.41	31.77	32	28	—
retail	0.03	fpgrowth	30	7.21	7.23	7.58	1.06	32	28	—
retail	0.03	eclat	30	2.39	2.37	2.55	9.24	32	28	—
retail	0.03	mlxtend	5	15.00	14.95	15.34	2667.7	32	28	—
synthetic_dense_5000	0.3	apriori	5	13.67	13.62	14.23	4.93	1737	5262	—
synthetic_dense_5000	0.3	fpgrowth	5	55.95	57.17	58.87	16.32	1737	5262	—
synthetic_dense_5000	0.3	eclat	5	2.58	2.53	2.73	16.05	1737	5262	—
synthetic_dense_5000	0.3	mlxtend	5	0.551	0.529	0.603	380.1	1737	5262	—
synthetic_dense_5000	0.5	apriori	5	1.42	1.29	2.07	13.68	60	20	—
synthetic_dense_5000	0.5	fpgrowth	5	9.90	10.43	11.64	10.63	60	20	—
synthetic_dense_5000	0.5	eclat	5	0.547	0.545	0.590	9.27	60	20	—
synthetic_dense_5000	0.5	mlxtend	5	0.025	0.026	0.027	7.38	60	20	—
synthetic_sparse_2000	0.02	apriori	5	0.441	0.445	0.473	0.826	269	407	—
synthetic_sparse_2000	0.02	fpgrowth	5	0.325	0.277	0.544	12.73	269	407	—
synthetic_sparse_2000	0.02	eclat	5	0.085	0.089	0.097	0.264	269	407	—
synthetic_sparse_2000	0.02	mlxtend	5	0.086	0.090	0.100	3.46	269	407	—
synthetic_sparse_2000	0.05	apriori	5	0.190	0.195	0.202	0.974	77	118	—
synthetic_sparse_2000	0.05	fpgrowth	5	0.179	0.181	0.208	0.176	77	118	—
synthetic_sparse_2000	0.05	eclat	5	0.056	0.053	0.068	0.776	77	118	—
synthetic_sparse_2000	0.05	mlxtend	5	0.034	0.035	0.039	0.836	77	118	—
t10i4d100k	0.01	apriori	1	592.9	592.9	592.9	0.000	385	17	4 failed: timeout 600 s
t10i4d100k	0.01	fpgrowth	5	72.63	71.45	77.40	292.7	385	17	—
t10i4d100k	0.01	eclat	5	6.49	6.39	7.53	49.25	385	17	—
t10i4d100k	0.01	mlxtend	1	101.9	101.9	101.9	0.000	385	17	4 failed: MemoryError: Unable to allocate 13.1 GiB
t10i4d100k	0.02	apriori	5	164.5	169.7	170.0	0.824	155	0	—
t10i4d100k	0.02	fpgrowth	5	48.41	48.07	49.81	124.5	155	0	—
t10i4d100k	0.02	eclat	5	4.78	4.80	5.01	23.61	155	0	—
t10i4d100k	0.02	mlxtend	5	34.40	40.95	42.95	17.31	155	0	—
tic_tac_toe	0.3	apriori	30	0.069	0.071	0.085	0.038	25	10	—
tic_tac_toe	0.3	fpgrowth	30	0.102	0.100	0.138	0.253	25	10	—
tic_tac_toe	0.3	eclat	30	0.024	0.023	0.031	0.059	25	10	—
tic_tac_toe	0.3	mlxtend	5	0.012	0.013	0.015	0.020	25	10	—
tic_tac_toe	0.5	apriori	30	0.046	0.046	0.058	0.025	1	0	—
tic_tac_toe	0.5	fpgrowth	30	0.045	0.044	0.064	0.021	1	0	—
tic_tac_toe	0.5	eclat	30	0.022	0.020	0.030	0.026	1	0	—
tic_tac_toe	0.5	mlxtend	5	0.005	0.005	0.006	0.040	1	0	—